Course Title: Research Methods and Pharmacoepidemiology

Course Description:

This course provides an integrated approach to pharmacoepidemiology, clinical trials, epidemiological measures, experimental design, systematic reviews, meta-analyses, and pharmacoeconomics. It includes modern topics such as Real-World Evidence, Drug Safety, Personalized Medicine, Antimicrobial Resistance, Vaccine Epidemiology, Global Health, and Regulatory Science, equipping pharmacy students with the essential knowledge and skills needed for research and practice.

Course Objectives:

- Master the principles and applications of pharmacoepidemiology and pharmacoeconomics.
- Learn how to conduct systematic reviews and meta-analyses.
- Understand the integration of real-world evidence and big data analytics.
- Develop skills in drug safety, pharmacovigilance, and personalized medicine.
- Explore global health issues and regulatory science in pharmacoepidemiology.

Course Modules & Lecture Breakdown:

1. **Lecture 1: Introduction to Pharmacoepidemiology and Research Methods**
    - Overview of pharmacoepidemiology and its significance in pharmacy practice.
    - Introduction to research methods relevant to drug research and public health.
2. **Lecture 2: Clinical Trial Design and Implementation**
    - Phases of clinical trials, adaptive designs, and decentralized trials.
    - Ethical considerations and regulatory requirements in clinical trials.
3. **Lecture 3: Epidemiological Measures and Data Analysis**
    - Key epidemiological measures and statistical methods in pharmacoepidemiology.
    - Real-world data applications and analysis techniques.
4. **Lecture 4: Real-World Evidence (RWE) and Big Data Analytics**
    - Integration of RWE from electronic health records, insurance claims, and patient registries.
    - Utilizing big data analytics and machine learning for drug research.
5. **Lecture 5: Drug Safety and Pharmacovigilance**
    - Principles of pharmacovigilance and adverse drug reaction (ADR) monitoring.

- Advanced techniques in ADR detection, including data mining and social media monitoring.

6. **Lecture 6: Personalized Medicine and Pharmacoepidemiology**
    - Advances in genomics and molecular biology for personalized treatment plans.
    - Impact of genetic variations on drug response and safety.
7. **Lecture 7: Experimental Design in Pharmacoepidemiology**
    - Principles of experimental design and advanced statistical models.
    - Application of experimental design in real-world research.
8. **Lecture 8: Systematic Reviews and Meta-Analyses**
    - Methodology and significance of systematic reviews and meta-analyses in pharmacoepidemiology.
    - Conducting and interpreting systematic reviews and meta-analyses.
9. **Lecture 9: Antimicrobial Resistance (AMR)**
    - Role of pharmacoepidemiology in tracking and combating AMR.
    - Global strategies and patterns of antibiotic use and resistance.
10. **Lecture 10: Vaccine Epidemiology**
    - Principles of vaccine research and monitoring during pandemics.
    - Evaluating vaccine effectiveness, safety, and real-world application.
11. **Lecture 11: Pharmacoeconomics: Cost-Minimization and Cost-Effectiveness Analyses**
    - Introduction to pharmacoeconomics and its role in healthcare decision-making.
    - Conducting cost-minimization and cost-effectiveness analyses in drug research.
12. **Lecture 12: Global Health and Pharmacoepidemiology**
    - Application of pharmacoepidemiological methods to global health challenges.
    - Understanding the impact of socioeconomic factors on drug use and health outcomes.
13. **Lecture 13: Regulatory Science and Post-Marketing Surveillance**
    - The role of pharmacoepidemiology in drug approval processes.
    - Post-marketing surveillance, regulatory requirements, and the use of RWE.
14. **Lecture 14: Practical Applications in Pharmacoepidemiology**

- Designing a comprehensive research project integrating clinical trials, RWE, and pharmacoeconomics.
- Case studies and real-world examples in drug safety, personalized medicine, and vaccine epidemiology.

15. **Lecture 15: Emerging Trends and Future Directions in Pharmacoepidemiology**
    - Innovations and future directions in pharmacoepidemiology and drug research.
    - Challenges and opportunities in integrating new technologies and methodologies.

Assessment Methods:

- **Quizzes and Exams:** Evaluate theoretical knowledge and understanding of key concepts.
- **Research Project:** Design and present a study integrating principles from clinical trials, epidemiological measures, systematic reviews, meta-analyses, and pharmacoeconomics.
- **Case Study Analysis:** Critique real-world studies and data, including drug safety and vaccine efficacy.

Course Materials:

- **Textbooks and Articles:** Core readings on pharmacoepidemiology, clinical trial design, RWE, drug safety, and personalized medicine.
- **Software Tools:** Training in statistical software for data analysis and big data analytics.
- **Case Studies:** Real-world examples from recent research on AMR, vaccine epidemiology, and global health.

Learning Outcomes:

Upon completion, students will:

- Have a comprehensive understanding of research methods, pharmacoepidemiology, and pharmacoeconomics.
- Be proficient in conducting systematic reviews, meta-analyses, and cost-effectiveness analyses.
- Develop skills in applying RWE, big data analytics, and personalized medicine to clinical practice.
- Be prepared to contribute to global health efforts, regulatory science, and patient safety.